TECHNICAL TRAINING PROGRAM

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Training Days 2026 – File System

File System & Storage Management

FILE SYSTEM – THE DIGITAL LIBRARY

A file system is a method used by an Operating System to store, organize, and manage files and directories on a storage device. It acts as a link between the Operating System and the physical storage hardware, allowing users and applications to create, read, update, and delete files in an organized way.

A file system can be compared to a library containing thousands of books. Without an organized system, finding a specific book would be difficult and time-consuming. In a similar way, the file system helps the computer store information in known locations and retrieve it when required.

Files – the books

Files are like the books in a library. Each file contains data, such as a document, image, video, audio recording, configuration file, or program.

A file normally has information associated with it, including:

- a name
- a file type or extension
- a size
- a location
- timestamps
- access permissions

Directories – the shelves

Directories, also called folders in many graphical interfaces, are like the shelves and sections of a library. They organize files into a hierarchical structure, making navigation, management, and searching easier.

A directory may contain files and other directories, which are called subdirectories or subfolders.

Space Management – the librarian

The librarian manages the available space and decides where books should be placed. Similarly, the file system allocates storage space to files and directories, keeps track of used and free space, and manages how data is placed on the storage device.

Depending on the file system and storage technology, files may become fragmented, meaning that parts of the same file are stored in different locations.

File Naming – the library catalog

A library catalogue records the names and locations of books. In the same way, a file system uses file and directory names to help users and applications identify and locate stored information.

File-naming rules depend on the Operating System and file system. Certain characters, name lengths, or reserved words may not be permitted.

Access Control – the locks

A library may restrict access to valuable or confidential material. Similarly, a file system can use permissions to control which users or groups may access specific files and directories.

Depending on the file system, permissions may control actions such as:

- reading
- creating
- modifying
- deleting
- executing files

Data integrity – the maintenance

The library must protect books from damage and ensure that their contents remain reliable. In a similar way, file systems use structures and mechanisms that help store and retrieve data correctly.

Some file systems include additional integrity or recovery features, but no file system can completely prevent data loss caused by hardware failure, malware, incorrect actions, or the absence of backups.

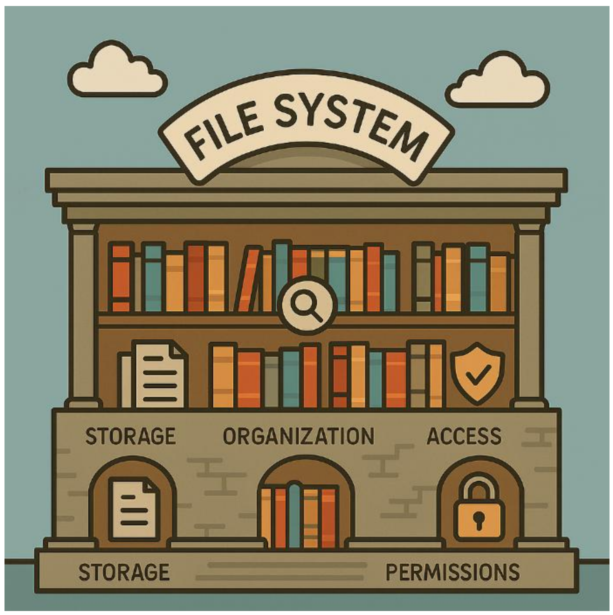
Metadata management – the notes

A library catalogue stores information about each book, such as its author, category, location, and publication date. This information is called metadata.

In a file system, metadata is data about a file or directory. It may include:

- file name
- size
- type
- location
- owner
- permissions
- timestamps

Metadata helps the Operating System and users organize, search for, manage, and protect files.



File System – Our Digital Library

OS SECURITY FEATURES AND MECHANISMS

OS Security

To understand the security features and mechanisms of an Operating System, imagine the computer as a castle containing valuable information and resources that must be protected from unauthorized access and other threats.

Access Control – "The Guards"

The castle guards control who may enter and which areas each person may access. In an Operating System, access control determines which users and processes may use specific files, applications, devices, and system resources.

Access control may use:

- user accounts and groups
- authentication
- permissions
- roles
- Access Control Lists (ACLs)

These mechanisms help ensure that users receive only the access required for their responsibilities.

Encryption – "The Keys"

A person with the correct key can unlock a protected area of the castle. In IT, encryption converts readable data into a protected form that cannot normally be understood without the correct decryption key.

Encryption may protect:

1. Data at Rest: Data stored on hard drives, SSDs, removable devices, or other storage systems.
2. Data in Transit: Data travelling across a network or the Internet.

Full-disk encryption is one example of protecting data at rest. It helps protect stored information if a device is lost or stolen.

Secure Boot – "Secure Gate"

The secure gate checks visitors before allowing them to enter the castle. In a computer, Secure Boot checks the digital signatures of boot software during startup and allows trusted components to load.

This helps prevent unauthorized or modified boot software from running before the Operating System starts. Secure Boot improves startup protection, but it does not replace antivirus software, updates, or other security controls.

Intrusion Detection & Prevention – "The spies"

The spies observe activity inside and around the castle and report suspicious behavior. In IT, **Intrusion Detection Systems (IDS)** monitor events and network traffic for signs of attacks, misuse, or unusual activity.

An **Intrusion Prevention System (IPS)** performs similar monitoring but can also take automatic action, such as blocking suspicious traffic or stopping a detected attack.

In simple terms:

IDS: Detects and alerts.

IPS: Detects and can automatically block or prevent.

Authentication & Authorization – "The Key Keepers"

The key keepers verify the identity of visitors before allowing them to enter the castle. In a computer system, authentication verifies the identity of a user, device, or service. Authorization then determines which resources and actions the authenticated user is permitted to access.

In simple terms:

Authentication: Who are you?

Authorization: What are you allowed to do?

Network Security – "The Castle walls"

The castle walls help protect the castle and control how people enter or leave. In a computer environment, network security uses different controls to protect devices, systems, and data during communication.

Firewalls monitor and filter incoming and outgoing network traffic according to security rules. They can allow legitimate communication while blocking unwanted or unauthorized connections.

Virtual Private Networks (VPNs) create protected connections over public or untrusted networks. They use encryption and authentication to help protect data during transmission and to provide secure remote access.

Our Cyber Fortress